Bureau
International des
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Mesures

Le système international d'unités
The International System of Units

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Annexe 2
Annex 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2
Part 2

Krypton spectral lamp ()

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CCL-CCTF Frequency Standards Working Group



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Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second
Part 2: Krypton spectral lamp ()

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Appendix 2

1. CIPM recommended value

The value $\lambda = 605\ 780\ 210.3$ fm

with a relative expanded uncertainty $U = 3.9 \times 10^{-9}$, where $U = k u_c$ $(k = 3)$, u_c being the combined standard uncertainty, applies to the radiation emitted by a discharge lamp. The radiation of ^{86}Kr is obtained by means of a hot-cathode discharge lamp containing ^{86}Kr, of a purity not less than 99 %, in sufficient quantity to assure the presence of solid krypton at a temperature of 64 K, this lamp having a capillary with an inner diameter from 2 mm to 4 mm and a wall thickness of about 1 mm.

It is estimated that the wavelength of the radiation emitted by the positive column is equal, to within 1 part in 10^8, to the wavelength corresponding to the transition between the unperturbed levels, when the following conditions are satisfied:

- the capillary is observed end-on from the side closest to the anode;
- the lower part of the lamp, including the capillary, is immersed in a cold bath maintained at a temperature within one degree of the triple point of nitrogen;
- the current density in the capillary is (0.3 ± 0.1) A cm^{-2}.

2. Source data

Adopted value

$f = 494\ 886\ 516.4\ \ (6)$ MHz
$u_c/y = 1.3 \times 10^{-9}$

for which:
$\lambda = 605\ 780\ 210.3\ \ (.8)$ fm
$u_c/y = 1.3 \times 10^{-9}$

calculated from

f/ kHz	u_c/y	source data
494 886 516 422 kHz	1.3×10^{-9}	Section 2.1

Source data

2.1.

The CCDM 1982 [1], [2] gives

$f_{Kr}/f_i = 1.044\ 919\ 242\ 05$
$u_c/y = 1.3 \times 10^{-9}$,
using the recommended operation conditions [3].

Using the recommended value of the absorbing molecule $^{127}I_2$, a_{16} or f component, R(127) 11-5 transition (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1) one obtains

$f_i = 473\ 612\ 214\ 712$ kHz
$u_c/y = 2.2 \times 10^{-11}$,

which leads to

$f_{Kr} = 494\ 886\ 516\ 422$ kHz
$u_c/y = 1.3 \times 10^{-9}$.

^{86}Kr spectral lamp radiation, $5d_5 - 2p_{10}$ transition

References

[1] Documents Concerning the New Definition of the Metre, *Metrologia*, 1984, **19**, 163-178.

[2] *BIPM, Com. Cons. Déf. Mètre*, 1982, **7**, M58..

[3] *BIPM Proc. Verb. Com. Int. Poids et Mesures*, 1960, **28**, 71-72 and *BIPM Comptes Rendus 11e Conf. Gén. Poids et Mesures*, 1960, **85**.